

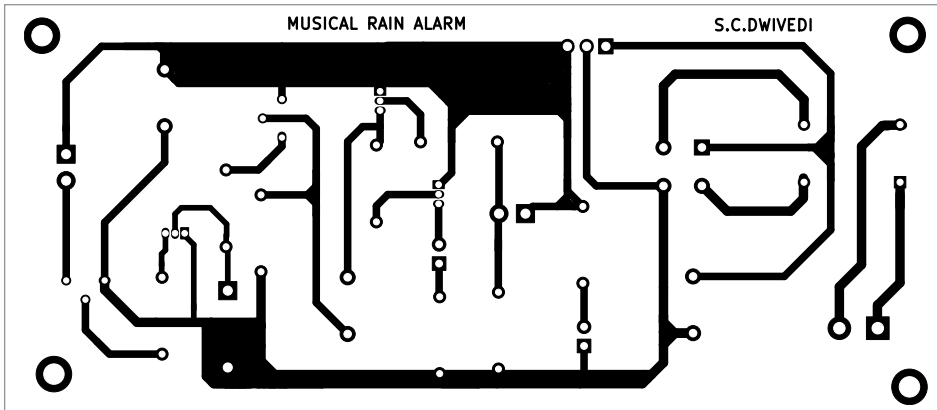


Fig. 4: Actual-size PCB

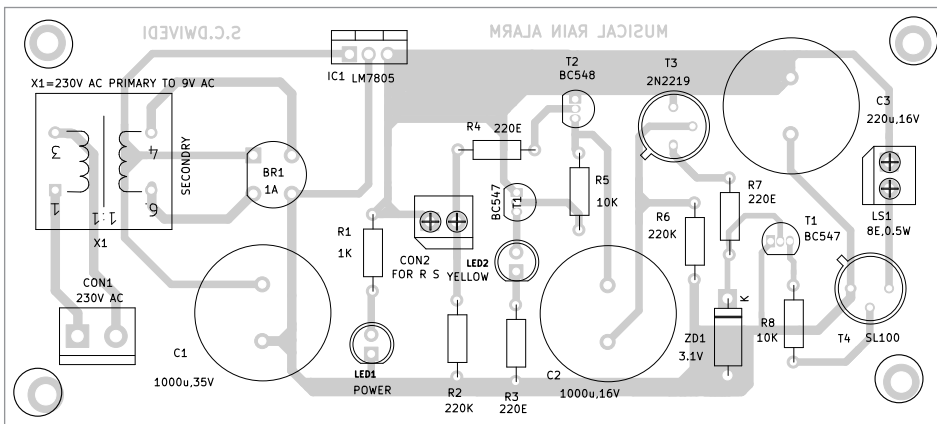


Fig. 5: Component side of the PCB

### PARTS LIST

#### Semiconductors:

|      |                                |
|------|--------------------------------|
| IC1  | - LM7805, 5V voltage regulator |
| IC2  | - UM66 melody generator        |
| T1   | - BC547 NPN transistor         |
| T2   | - BC548 NPN transistor         |
| T3   | - 2N2219 NPN transistor        |
| T4   | - SL100 NPN transistor         |
| BR1  | - 1A bridge rectifier          |
| LED1 | - 5mm red LED                  |
| LED2 | - 5mm yellow LED               |

#### Resistors (all 1/4-watt, $\pm 5\%$ carbon):

|            |                |
|------------|----------------|
| R1         | - 1-kilo-ohm   |
| R2         | - 220-kilo-ohm |
| R3, R4, R7 | - 220-ohm      |
| R5, R8     | - 10-kilo-ohm  |

#### Capacitors:

|    |                                  |
|----|----------------------------------|
| C1 | - 1000 $\mu$ F, 35V electrolytic |
| C2 | - 1000 $\mu$ F, 16V electrolytic |
| C3 | - 220 $\mu$ F, 16V electrolytic  |

#### Miscellaneous:

|            |                                                      |
|------------|------------------------------------------------------|
| CON1, CON2 | - 2-pin connector                                    |
| LS1        | - 8 $\Omega$ , 0.5-watt speaker                      |
| X1         | - 230V AC primary to 9V, 500mA secondary transformer |
| RS         | - Rain sensor                                        |

heart of this device is a rain sensor connected across CON2. The circuit comprises step-down transformer X1, bridge rectifier BR1, 5V voltage regulator LM7805 (IC1), rain sensor (RS), four NPN transistors (T1 through T4), melody generator UM66 (IC2), speaker (LS1), 3.1V zener diode, and a few other components.

This device operates with 5V DC, which is derived from the 230V AC primary to 9V, 500mA step-down transformer X1. The 230V AC mains is connected to the primary of X1 via CON1 in the circuit. The secondary of X1 is at 9V AC, which is connected to the bridge rectifier BR1 for rectification and filtered by capacitor C1. The rectified 5V is given to the rain sensor and the circuit to operate.

In this circuit, we are interfacing

a musical rain sensor with the melody IC UM66. We have used two LEDs in the circuit; one (LED1) is for the power supply indicator, and the second (LED2) is yellow, indicating that rain is continuing. Fig. 3 shows the image of the rain sensor used here. It has two terminals which need to connect across CON2 during installation.

When the rain starts, the sensor provides positive voltage at point 'B', which makes transistor T2 to conduct. The capacitor C2 then charges and transistor T3 conducts; IC2 gets 3.1V to enable the melody generator sound via transistor T4 and the speaker. At the same time, transistor T1 also conducts, and the yellow LED glows to indicate the status of rain.

### Construction and testing

An actual-size, single-side PCB layout for the musical rain alarm is shown in Fig. 4,

and its component layout is shown in Fig. 5. After assembling the circuit on PCB, enclose it in a suitable box. Fix LED1, LED2, and CON2 on the front side of the box and the mains power connector CON1 on the rear side of the box.

Connect two pins of the rain sensor with twin wire and install it on the roof in such a way that raindrops should fall on it. Connect the other ends of the twin wires to CON2 on the PCB. **EFY**

**Bonus.** You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/live-diymusical-rain-alarm>

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